

INDEPENDENT PROTOTYPE · SYNTHETIC DATA

VC Retail Analytics

An independent analytics prototype for designer-lighting retail workflows.
Synthetic retail data, controlled metric queries, and a deterministic assistant
for exploring margin, seasonality, field performance, and supply.

The assistant is **not an LLM**. It routes exact supported prompts to approved metric queries
and returns traceable SQL and rows. That constraint is the engineering point: tool access,
metric definitions, user scope, and testable responses.

Open hosted demo →

<https://vc.datasharkbi.com> [GitHub \(public\)](#)

SYNTHETIC

SQLITE ON FLY

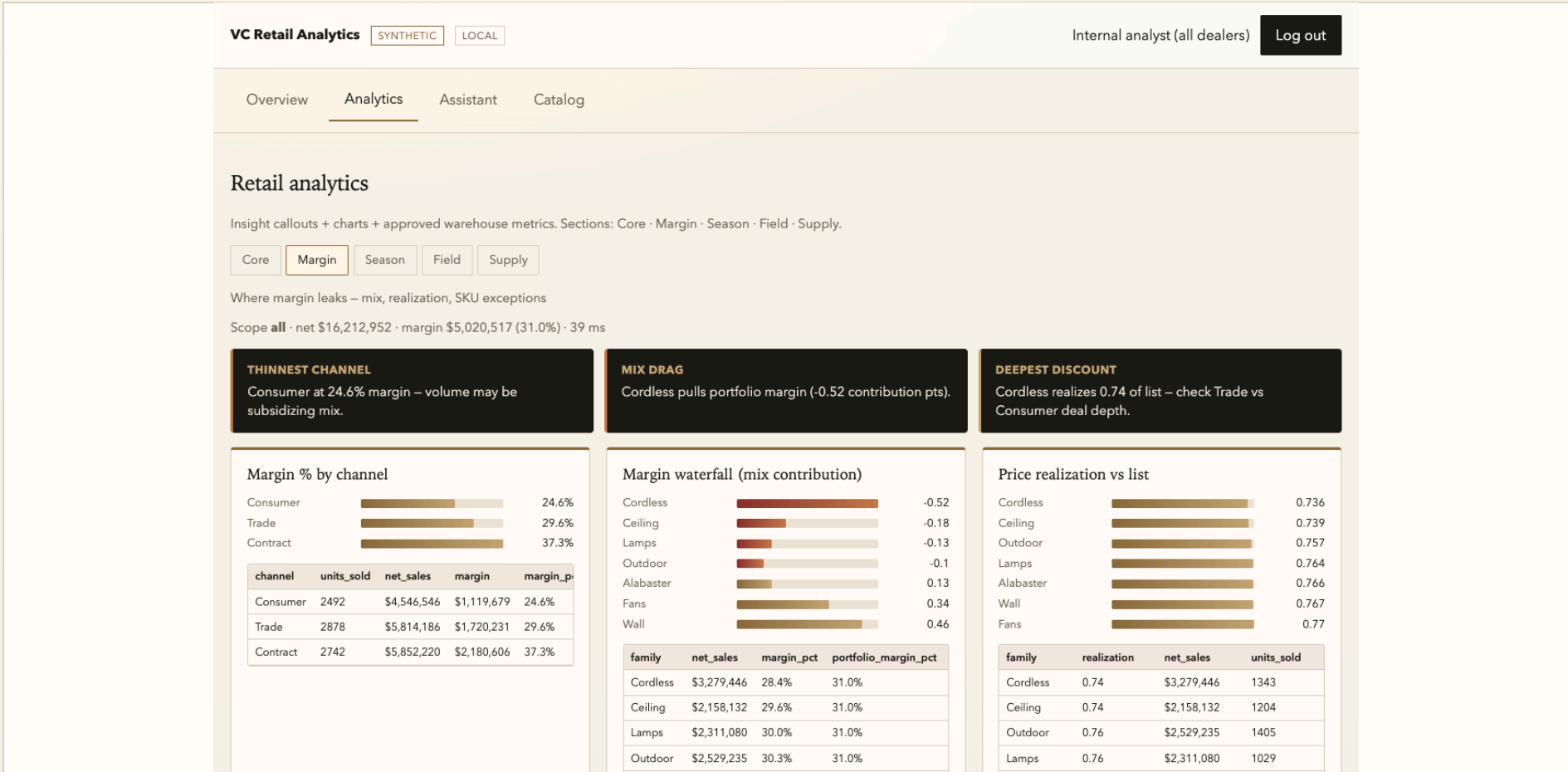
DETERMINISTIC ROUTER

ACCESS-GATED

Paul Rendleman · independent prototype

Which channel has the thinnest margin?

Live capture from the synthetic app (operator session). Margin section shows margin % by channel, mix contribution waterfall, and price realization. Insight callouts summarize the approved metric results.



Supported prompt → approved query → result

VC Retail Analytics

SYNTHETICLOCAL

Internal analyst (all dealers)Log out

OverviewAnalyticsAssistantCatalog

Illustrative support vs analytics workflow

Left: illustrative FAQ / escalate pattern. Right: deterministic assistant that routes exact prompts to approved metric queries (not an LLM). Illustrative workflows, not an assessment of any client's internal systems.

Illustrative support workflow

e.g. custom order lead time

Ask support pattern

Analytics workflow (demonstrated)

show margin percent

Ask approved metricEscalate packet

offline governed assistant (not LLM) - margin_pct

Thinnest channel in result set: Consumer at 24.6% margin.

Margin percent by channel (margin \$ / net sales).

Consumer24.6%

Trade29.6%

Contract37.3%

channel	units_sold	net_sales	margin	margin_pct
Consumer	2492	\$4,546,546	\$1,119,679	24.6%
Trade	2878	\$5,814,186	\$1,720,231	29.6%
Contract	2742	\$5,852,220	\$2,180,606	37.3%

► SQL + tool trace

► Supported metric prompts (exact match)

Exact prompt used

show margin percent

What returned

Metric margin_pct, one-line insight, bar chart, table, and expandable SQL + tool trace. Scope follows the signed session (operator = all dealers).

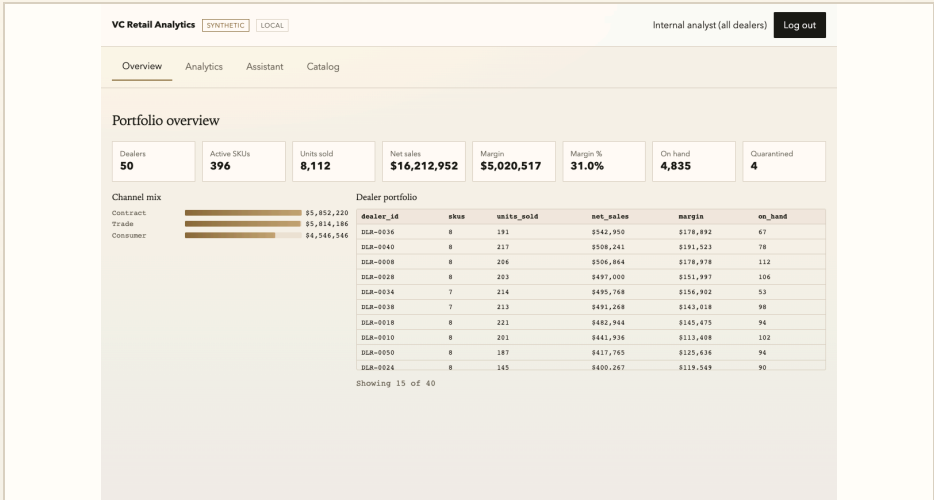
Unsupported requests

Invented forecasts and free-form questions are refused or clarified. The router only accepts an exact supported vocabulary.

Proposed (not built):

an LLM could interpret natural language, then call the same approved tools, with authorization enforced outside the model.

Analytics sections actually implemented



Illustrative rule-based scores

Rep A-D grades and vendor Prefer / Watch / Exit are synthetic, rule-based composites (attainment, margin vs peer, mix, OTIF, quality). They are demo decision aids, not real personnel or supplier recommendations.

Core

Channel, family, region sell-through and stock-risk flags.

Margin

Margin %, realization, mix waterfall, low-margin SKUs.

Season

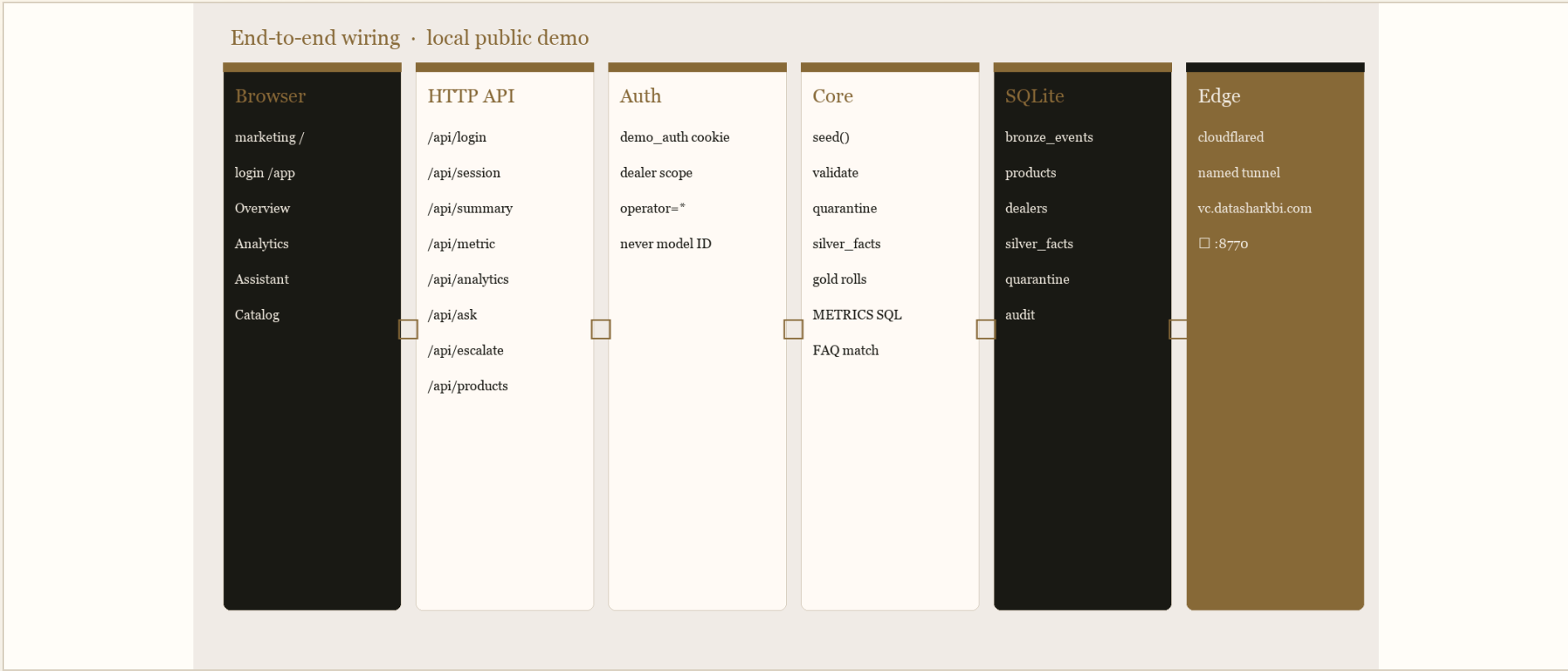
Seasonal index heatmap, YoY, lead-vs-peak buy-ahead flags.

Field & Supply

Territories, whitespace, rep grades; days of cover, reorder, vendor scorecard.

Five Analytics subnav sections (Core, Margin, Season, Field, Supply) covering the six investigation themes listed in the repository README.

Browser → Access → hosted app → SQLite



Access gate

Cloudflare Access email OTP in front of the hostname.
Demo app login is separate (`operator` / `vc-demo`).

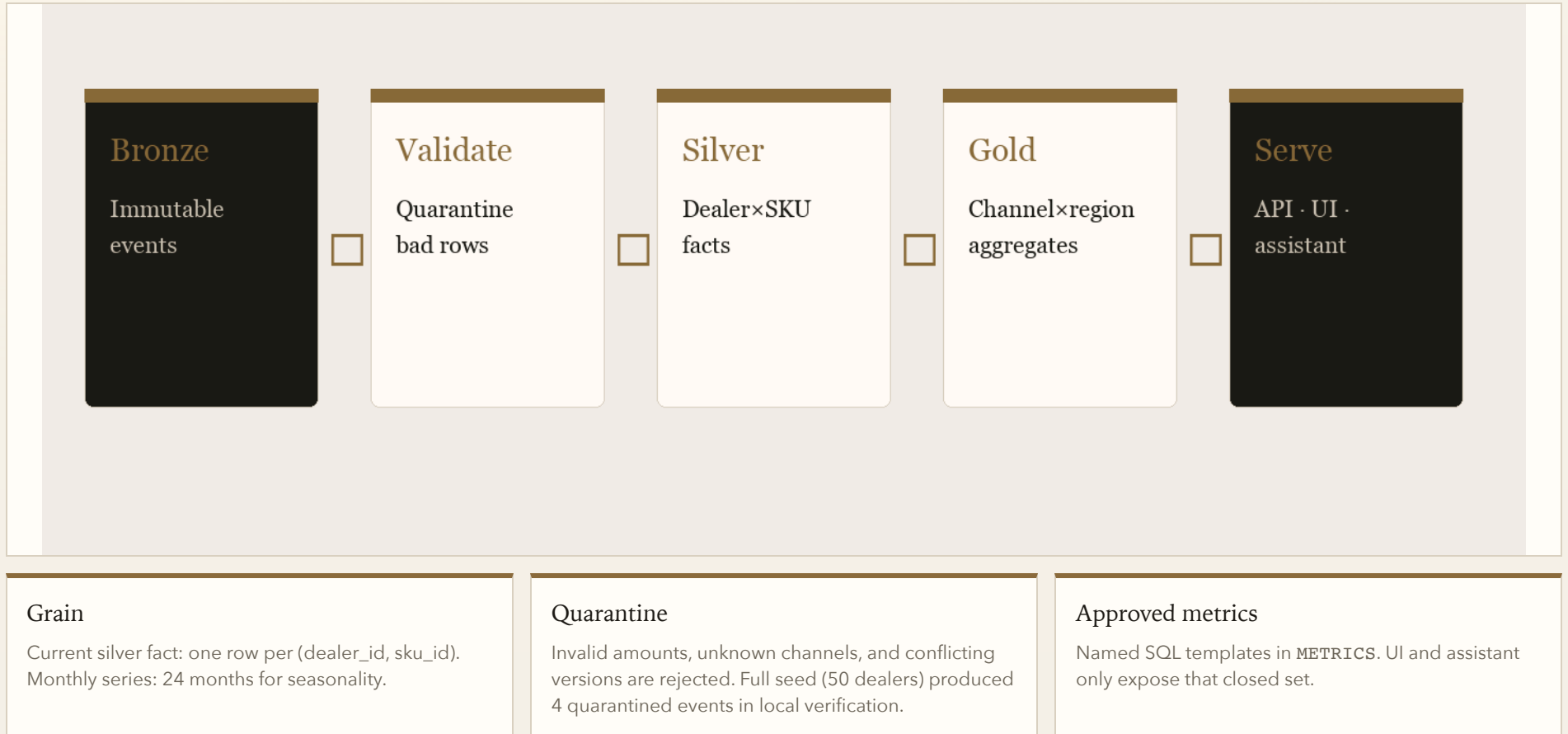
Hosted demo

Fly.io app `vc-retail-analytics` (ord) runs Python
+ tunnel. Observed machine state: started at
verification time.

Session scope

HMAC-signed cookie binds dealer or operator role.
Dealer sessions cannot request another dealer via
query or body.

| Sources → silver facts → served metrics



Engineering checks and observed results

Local checks: b40c9f6, 2026-10-02T20:38Z (re-run PASS). Hosted smoke: b40c9f6 image, 2026-10-02T20:45Z, PASS. Details in docs/VC_Submission_Verification.md.

CHECK	OBSERVED RESULT	EVIDENCE
Unit / API suite	PASS	6/6 tests in <code>tests/test_core.py</code> + <code>tests/test_assistant.py</code>
Dealer isolation	PASS	DLR-0001 session → GET summary?dealer=DLR-0002 returns 403; ask with other dealer denied; portfolio rows only DLR-0001
Auth failure	PASS	Wrong password → 403
Metric reconcile	PASS	API summary net/margin matched independent SQL on silver_facts (tolerance \$0.02)
Unsupported forecast	PASS	Local: 400 clarify. Hosted: HTTP 400 (no fabricated rows)
Quarantine seed	PASS	Full seed quarantined=4 invalid fixture events
Snowflake parity	Not run	Adapter and SQL exist; no reconcile evidence in this pass
Hosted authenticated walkthrough	PASS	Operator path on vc.datasharkbi.com: Overview, Analytics (5 sections), assistant metric + SQL trace, catalog search, logout

What is running vs proposed

COMPONENT	STATUS	NOTES
Hosted app + SQLite	Running	Fly machine started; Access OTP gate; authenticated hosted walkthrough PASS (see verification notes)
Deterministic assistant	Implemented + tested	Exact-match router; SQL trace; forecast refusal tested
Session dealer scope	Implemented + tested	Server-side scope + isolation tests
Analytics depth (5 sections)	Implemented	Margin / season / field / supply metrics in METRICS + UI
Snowflake adapter / RAP	Implemented, unverified live	Code + SQL present; not reconciled in this submission
LLM integration	Proposed	Not running; would call the same approved tools

Public support and authenticated dealer analytics serve different needs. This prototype explores how controlled warehouse metrics could support dealer and internal questions alongside familiar FAQ and escalation workflows. It is not an assessment of Visual Comfort internal systems.

| How to review the hosted demo

Access

- Open <https://vc.datasharkbi.com>
- Complete Cloudflare Access email OTP (allowlisted addresses only)
- Demo login: `operator / vc-demo`
- Optional tenancy check: `d1r-0001 / vc-demo`

Supported prompts to try

- `show margin percent`
- `show vendor scorecard`
- `show rep grades`
- `show lead vs peak`
- Refuse check: ask for an invented forecast

Fallback if Access blocks you

Screenshots in this PDF are from the current synthetic build. Source is public at <https://github.com/prendleman/vc-retail-analytics>. Local run: `python3 -m app.server` then `http://127.0.0.1:8770/login`. Additional reviewers may need access enabled before opening the hosted demo.

If an older UI still appears after a deploy, a normal refresh should pull versioned assets.

CLOSE

What this prototype demonstrates

Synthetic dealer analytics with approved metric definitions, session scope, quarantine, and a deterministic assistant that returns traceable results. Useful foundation for later LLM tooling without inventing answers.

Limitations: synthetic data only; Snowflake not live-reconciled here; Access OTP required for the hosted hostname; scores are illustrative rules.

Open hosted demo →

<https://vc.datasharkbi.com> github.com/prendleman/vc-retail-analytics

Paul Rendleman

| Snowflake path (code present, live not verified)

One UI can target SQLite or Snowflake. This submission’s hosted runtime is SQLite on Fly. Snowflake SQL, serving views, and RAP scripts exist under `sql/snowflake/` and `scripts/cloud.py`.

Present

Platform, transform, governance, semantic SQL; dual-backend app flag.

Not shown here

Live reconcile, RAP enforcement proof, Cortex Analyst evaluation.

Validation status

Live Snowflake validation pending. Backend reconciliation and row-access policy checks have not been demonstrated in this submission.

| Approved metric vocabulary (excerpt)

Core / margin / season

- portfolio, by_channel, by_family, by_region, stock_risk
- margin_pct, price_realization, low_margin_skus, margin_waterfall
- units_by_month, seasonal_index, yoy_family, lead_vs_peak

Field / supply / quality

- territory_perf, territory_coverage, whitespace
- rep_leaderboard, rep_grade
- days_of_cover, reorder_candidates, plan_vs_season
- vendor_otif, vendor_scorecard, quality

Full definitions live in `app/core.py` (`METRICS` / `METRIC_PHRASES`). Assistant accepts only exact phrases from that map.